



Encrypt Data with AWS KMS



Owais Yousuf

✖ **Access denied to kms:Decrypt**
You don't have permission to *kms:Decrypt*. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#) ↗

ser/nextwork-kms-user

Action: kms:Decrypt

On resource(s): arn:aws:kms:eu-north-1:197446685099:key/b41cf251-b0bc-46c1-9dfe-1de73272312e

Context: no identity-based policy allows the action



Diagnose with Amazon Q



Introducing Today's Project!

In this project, I will demonstrate how to create a KMS key, create DynamoDB and encrypt it with the KMS keys and validate the KMS encryption. The goal is to encrypt the database with the KMS encryption keys and safeguard it from being accessed unauthorisedly.

Tools and concepts

Services I used include KMS, IAM user and DynamoDB. Key concepts I learnt include encryption using KMS keys, creating IAM user and creating DynamoDB and encryption it using KMS key. Created IAM user and gave it permission to encrypt, decrypt database by modifying KMS key's policy. This project brought a great knowledge of KMS service, IAM service and DynamoDB service by AWS in me.

Project reflection

This project took me approximately 1 hour or less. There were no challenging part in it. It were an easy project and a fantastic one.

I chose to do this project today because of my SOC analyst journey. Something that would make learning with NextWork even better is that NextWork provide a great platform for projects and learning also.



Encryption and KMS

Encryption is the process of converting plaintext into jumbled text known as ciphertext which is non-readable by a normal human. Encryption provides security to the data it encrypts. Companies and developers do this to secure data and meet the compliance. Encryption keys are also ciphertext that are used to tell the encryption algorithm how to encrypt data and are also used to decrypt the data.

AWS KMS is a key management service by amazon. Key management systems are important because a key is a ciphertext which can't be remembered by the human so it needs to be stored safely.

Encryption keys are broadly categorized as symmetric and asymmetric. In symmetric key encryption single key is used for encryption and decryption while as in asymmetric two "public and private" keys are used for encryption. Public key for encryption and private key decryption. I set up a symmetric key because symmetric keys are faster and efficient for encrypting and decrypting large amount of data.



Success

Your AWS KMS key was created with alias `nextwork-kms-key` and key ID `b41cf251-b0bc-46c1-9dfe-1de73272312e`.

View key

Customer managed keys (1)

Key actions

Create key

Filter keys by properties or tags

< 1 >

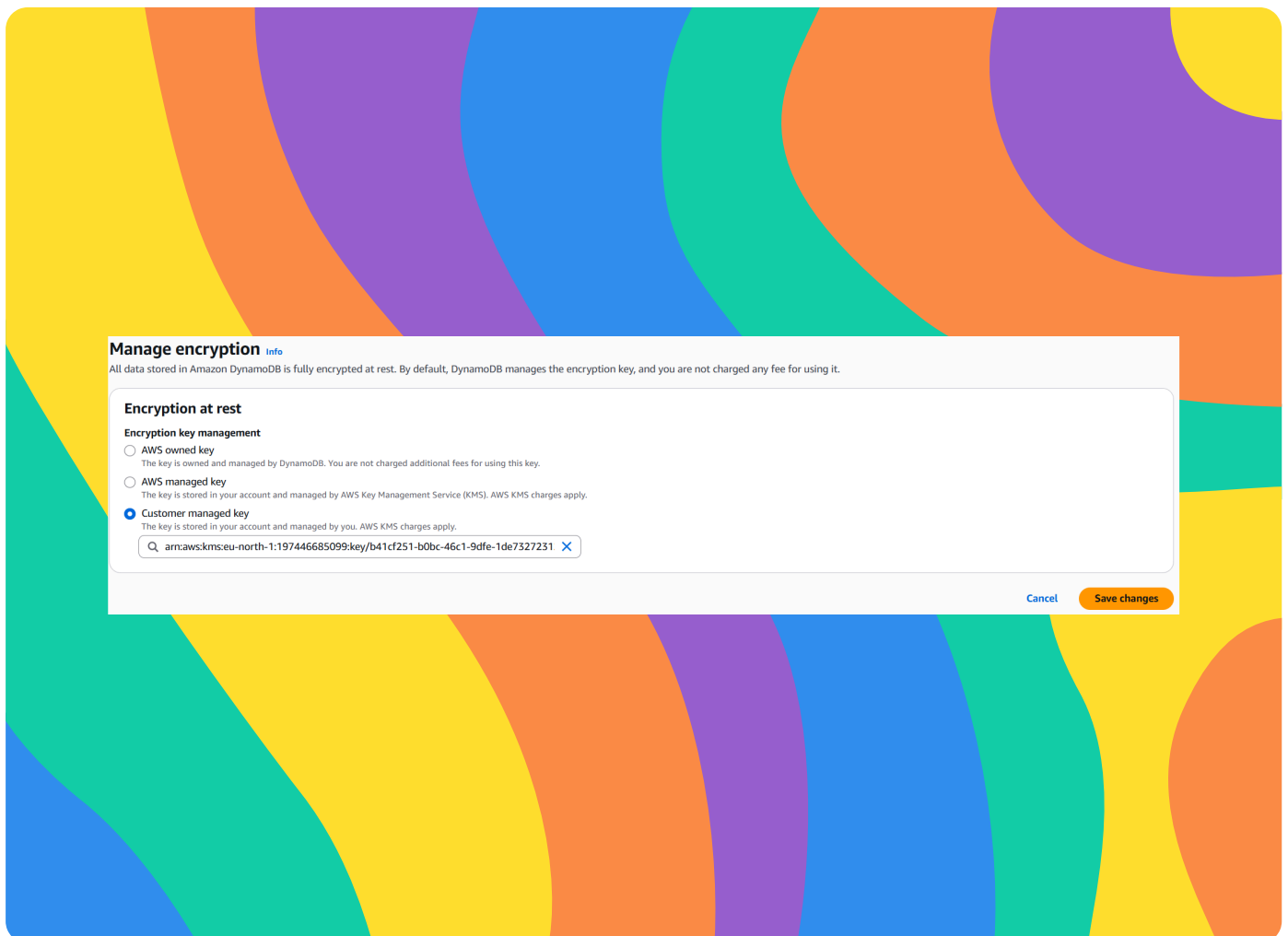
☐	Aliases	Key ID	Status	Key type	Key spec
☐	nextwork-kms-key	b41cf251-b0bc-46c1-9dfe-1...	Enabled	Symmetric	SYMMETRIC_DEFAULT



Encrypting Data

My encryption key will safeguard data in DynamoDB, which is a database service by AWS. DynamoDB is a fast and flexible database which makes to a good choice for stroing and retrieving large amount of data.

The different encryption options in DynamoDB include AWS owned key, AWS managed key and Customer managed key. Their differences are based on ownership and access. I selected Customer Managed Key(CMK) for the full control and access of key.

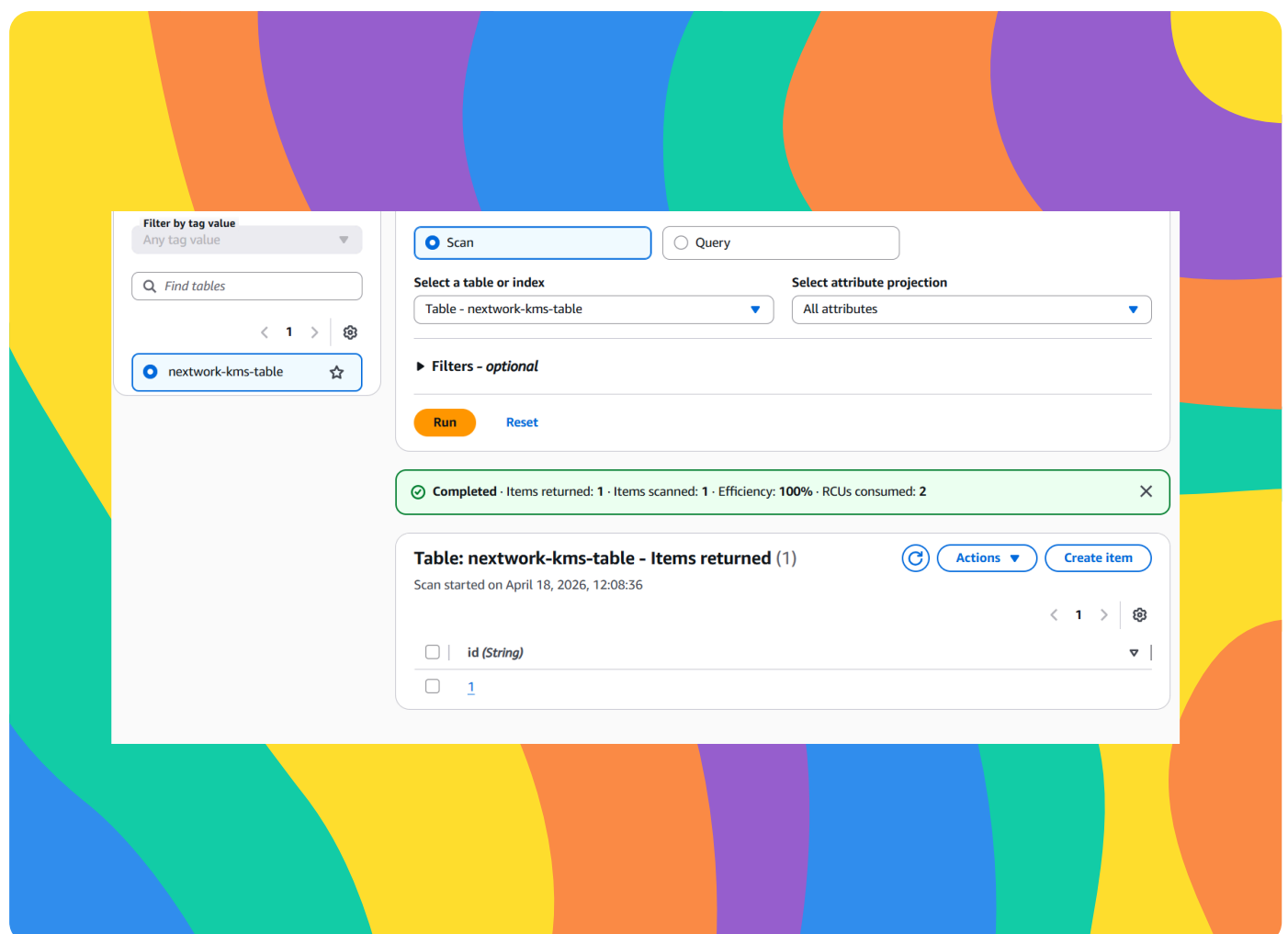




Data Visibility

Rather than controlling who has access to the key, KMS manages user permissions through resource-based key policies that explicitly define who can perform actions like encrypt, decrypt, or manage the key.

Despite encrypting my DynamoDB table, I could still see the table's items like id because of transparent data encryption. DynamoDB uses transparent data encryption, which means DynamoDB can decrypt the data on my behalf. Also when authorized user or application wants data DynamoDB retrieves the encrypted data, decrypts it with the key and show the decrypted data which make it fast.



The screenshot displays the AWS IAM console interface for a scan operation. The background features a colorful abstract pattern. The interface includes a sidebar with a search bar and a list of tables, with 'nextwork-kms-table' selected. The main panel shows the 'Scan' operation configuration, including a table selector and an attribute projection dropdown. A status bar indicates the scan is 'Completed' with 1 item returned and 100% efficiency. Below this, a table titled 'Table: nextwork-kms-table - Items returned (1)' shows a single item with the attribute 'id' having the value '1'.

id (String)
1



Denying Access

I configured a new IAM user to test the KMS encryption. The permission policies I granted this user are "AmazonDynamoDBFullAccess" which will grant full access to the DynamoDB.

After accessing the DynamoDB table as the test user, I encountered an error message that says "Access denied to kms:Decrypt" because the test user doesn't have permission of kms decrypt that's why this error occurred. This confirmed the validation of KMS encryption to the DynamoDB.

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EXTRA: Granting Access

To let my test user use the encryption key, I change the key policy in the KMS customer managed keys section. My key's policy was updated to do the actions such as encrypt, decrypt, ReEncrypt, GenerateDataKey and DescribeKey.

Using the test user, I retried the DynamoDB by refreshing the tab using the test user. I observed that I now have permission to decrypt and view the the database and the content of it which confirmed that the test user has now permission of decryption.

Encryption secures data within the database which add extra layer of security to the data in database. The other access control tools like security groups or permission policies control access to resources only not to the data inside.

```
38 },
39 {
40   "Sid": "Allow use of the key",
41   "Effect": "Allow",
42   "Principal": {
43     "AWS": [
44       "arn:aws:iam::197446685099:user/owais-yousuf-IAM",
45       "arn:aws:iam::197446685099:user/nextwork-kms-user"
46     ]
47   },
48   "Action": [
49     "kms:Encrypt",
50     "kms:Decrypt",
51     "kms:ReEncrypt*",
52     "kms:GenerateDataKey*",
53     "kms:DescribeKey"
54   ],
55   "Resource": "*"
56 }
```



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